

DBMS – SELLER AND INVENTORY MANAGEMENT

Project Title : SHOPWITHNYKAA

1. Database Design

1.1 Seller Table — Attributes

Attribute	Key / Constraint
Seller_ID	Primary Key
Seller_Name	NOT NULL
Email	UNIQUE
Phone	—
Address	—
City	—
Status	—

1.2 Inventory Table — Attributes

Attribute	Key / Constraint
Inventory_ID	Primary Key
Product_ID	Foreign Key
Seller_ID	Foreign Key
Quantity	—
Reorder_Level	—
Last_Updated	—
Inventory_Status	—

2. Table Relationships

- Seller.Seller_ID → Primary Key
- Inventory.Inventory_ID → Primary Key
- Inventory.Seller_ID → Foreign Key referencing Seller.SellerID

Relationship: One-to-Many (1:M) — one seller can have multiple inventory records.

2. Table Creation

2.1 Create Seller Table

```
SQL> CREATE TABLE Seller (  
2     Seller_ID NUMBER PRIMARY KEY,  
3     Seller_Name VARCHAR2(100) NOT NULL,  
4     Email VARCHAR2(100) NOT NULL UNIQUE,  
5     Contact_No VARCHAR2(15) NOT NULL UNIQUE,  
6     Address VARCHAR2(200) NOT NULL  
7 );  
  
Table created.
```

2.2 Create Inventory Table

```
SQL> CREATE TABLE Inventory (  
2     Inventory_ID NUMBER PRIMARY KEY,  
3     Product_ID NUMBER,  
4     Seller_ID NUMBER,  
5     Stock_Quantity NUMBER NOT NULL,  
6     Stock_Status VARCHAR2(20) NOT NULL,  
7     Last_Updated DATE NOT NULL,  
8     CONSTRAINT fk_inventory_seller  
9         FOREIGN KEY (Seller_ID)  
10        REFERENCES Seller(Seller_ID)  
11 );  
  
Table created.
```

3 .Seller Data Insertion

```
SQL> INSERT INTO Seller  
2 VALUES (101, 'Green Earth Store', 'greeneearth@gmail.com', '9876543210', 'Chennai');  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller  
2 VALUES (102, 'Eco Mart', 'ecomart@gmail.com', '9876543211', 'Bangalore');  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller  
2 VALUES (103, 'Nature Hub', 'naturehub@gmail.com', '9876543212', 'Hyderabad');  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller  
2 VALUES (104, 'Organic World', 'organicworld@gmail.com', '9876543213', 'Mumbai');  
1 row created.  
  
SQL>  
SQL> INSERT INTO Seller  
2 VALUES (105, 'Eco Life', 'ecolife@gmail.com', '9876543214', 'Delhi');  
1 row created.
```

4. Inventory Data Insertion

```
SQL> SELECT * FROM Inventory;  
  
INVENTORY_ID PRODUCT_ID SELLER_ID STOCK_QUANTITY STOCK_STATUS LAST_UPDA  
-----  
201          1001         101           50 Available 17-SEP-26  
202          1002         102           30 Available 17-SEP-26  
203          1003         103            0 Unavailable 17-SEP-26  
204          1004         104           25 Available 17-SEP-26  
205          1005         105            0 Unavailable 17-SEP-26
```

5. Seller Product Information

5.1 Display Seller and Product Details

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     s.Email,
  5     i.Product_ID,
  6     i.Stock_Quantity,
  7     i.Stock_Status
  8 FROM Seller s
  9 JOIN Inventory i
 10 ON s.Seller_ID = i.Seller_ID;
```

SELLER_ID		
SELLER_NAME		
EMAIL		
PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
101		
Green Earth Store		
greeneearth@gmail.com		
1001	50	Available

SELLER_ID		
SELLER_NAME		
EMAIL		
PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
102		
Eco Mart		
ecomart@gmail.com		
1002	30	Available

SELLER_ID		
SELLER_NAME		
EMAIL		
PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
103		
Nature Hub		
naturehub@gmail.com		
1003	0	Unavailable

SELLER_ID		
SELLER_NAME		
EMAIL		
PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
104		
Organic World		
organicworld@gmail.com		
1004	25	Available

SELLER_ID		
SELLER_NAME		
EMAIL		
PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
105		
Eco Life		
ecolife@gmail.com		
1005	0	Unavailable

6. Inventory Status

6.1 Display Available Products

```
SQL> SELECT
  2     s.Seller_Name,
  3     i.Product_ID,
  4     i.Stock_Quantity,
  5     i.Stock_Status
  6 FROM Seller s
  7 JOIN Inventory i
  8 ON s.Seller_ID = i.Seller_ID
  9 WHERE i.Stock_Status = 'Available';
```

SELLER_NAME	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
Green Earth Store	1001	50	Available
Eco Mart	1002	30	Available
Organic World	1004	25	Available

6.2 Display Unavailable Products

```
SQL> SELECT
  2     s.Seller_Name,
  3     i.Product_ID,
  4     i.Stock_Quantity,
  5     i.Stock_Status
  6 FROM Seller s
  7 JOIN Inventory i
  8 ON s.Seller_ID = i.Seller_ID
  9 WHERE i.Stock_Status = 'Unavailable';
```

SELLER_NAME	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS
Nature Hub	1003	0	Unavailable
Eco Life	1005	0	Unavailable

7. Inventory Update

```
SQL> UPDATE Inventory
  2 SET Stock_Quantity = 75,
  3     Stock_Status = 'Available',
  4     Last_Updated = SYSDATE
  5 WHERE Inventory_ID = 201;
```

1 row updated.

8.Inventory Status Reports

8.1 Complete Inventory Report

```
SQL> SELECT
  2     s.Seller_Name,
  3     i.Product_ID,
  4     i.Stock_Quantity,
  5     i.Stock_Status,
  6     i.Last_Updated
  7 FROM Seller s
  8 JOIN Inventory i
  9 ON s.Seller_ID = i.Seller_ID;
```

SELLER_NAME	PRODUCT_ID	STOCK_QUANTITY	STOCK_STATUS	LAST_UPDA
Green Earth Store	1001	75	Available	17-SEP-26
Eco Mart	1002	30	Available	17-SEP-26
Nature Hub	1003	0	Unavailable	17-SEP-26
Organic World	1004	25	Available	17-SEP-26
Eco Life	1005	0	Unavailable	17-SEP-26

8.2 Available and Unavailable Product Count

```
SQL> SELECT
  2     Stock_Status,
  3     COUNT(*) AS Product_Count
  4 FROM Inventory
  5 GROUP BY Stock_Status;
```

STOCK_STATUS	PRODUCT_COUNT
Available	3
Unavailable	2

```

SQL> SELECT
  2     s.Seller_Name,
  3     SUM(i.Stock_Quantity) AS Total_Stock
  4 FROM Seller s
  5 JOIN Inventory i
  6 ON s.Seller_ID = i.Seller_ID
  7 GROUP BY s.Seller_Name;

```

SELLER_NAME	TOTAL_STOCK
Green Earth Store	75
Eco Mart	30
Nature Hub	0
Organic World	25
Eco Life	0

8.3 Total Stock by Seller

```

SQL> SELECT
  2     s.Seller_Name,
  3     SUM(i.Stock_Quantity) AS Total_Stock
  4 FROM Seller s
  5 JOIN Inventory i
  6 ON s.Seller_ID = i.Seller_ID
  7 GROUP BY s.Seller_Name;

```

SELLER_NAME	TOTAL_STOCK
Green Earth Store	75
Eco Mart	30
Nature Hub	0
Organic World	25
Eco Life	0

8.4 Products with Zero Stock

```
SQL> SELECT
  2     Product_ID,
  3     Seller_ID,
  4     Stock_Quantity,
  5     Stock_Status
  6 FROM Inventory
  7 WHERE Stock_Quantity = 0;
```

PRODUCT_ID	SELLER_ID	STOCK_QUANTITY	STOCK_STATUS
1003	103	0	Unavailable
1005	105	0	Unavailable

